

Permanent and Transitory Income Effects in a Model of Optimal Consumption with Wage Income Uncertainty*

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I. INTRODUCTION

In this paper we solve a model of optimal consumption with wage uncertainty under two special assumptions about the form of the uncertainty. First, wages are governed by the process

$$w_t = \rho w_{t-1} + \epsilon_t, \quad (1)$$

where w_t is the wage in time t , ρ is constant and ϵ_t is an identically, independently distributed (i.i.d.) random variable;

$$\epsilon_{\min} \leq \epsilon_t \leq \epsilon_{\max},$$

where $\epsilon_{\max} \leq \infty$. Second, the distribution of ϵ_t comes from a location and scale family, which we define below.

Certainly part of the interest in looking at this kind of model comes from the fact that the scenario underlying it, multiperiod planning with a random wage, seems to be the same as that underlying Friedman's Permanent Income Hypothesis (PIH).¹ That being the case, when we derive the optimal consumption function, will it be consistent with the PIH? Although the latter is consistent with a good deal of the data relating consumption and various concepts of income and wealth, surely an important prior question is whether it holds together theoretically. However, except for a paper by Yaari [14], nobody seems to have looked at this question in the context of a stochastic, dynamic model close to the one Friedman had in mind.

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¹ See Friedman [5].

We will find that if $\rho = 0$, so that the wage is i.i.d., the location parameter of the wage distribution can be thought of as permanent income, and enters the optimal consumption function in a way consistent with the PIH. However, if $\rho \neq 0$, all changes in the wage are permanent. In this case a reformulated version of the PIH comes out of the optimal consumption function.

A second point of interest in this type of model involves the effects on optimal savings of increased riskiness in the wage distribution. We extend a two-period result due to Leland to the multiperiod case: that increased wage uncertainty raises or lowers saving according to whether the third derivative of the utility function is positive or negative. Since the plausible requirement that the consumer's utility function display decreasing absolute risk aversion implies a positive third derivative, this establishes a presumption that optimal saving rise with wage uncertainty.

II. STATEMENT OF THE PROBLEM

The consumer acts so as to maximize the expected sum of discounted utilities of consumption over a planning horizon of T periods. His utility function is increasing, concave, and thrice continuously differentiable. In each time period he receives a wage which follows the process (1). In any time period t , given the outcome of the wage and given net worth, the consumer can invest in an asset with a safe return r and can also borrow at that rate. Principal plus interest on a loan must be repaid in the period immediately following the one in which the debt was incurred. Since 'death' occurs after period T , the consumer cannot borrow in that period.

We begin with notation:

$$w_t = \text{wage in time } t, \quad i = 0, 1, \dots, t, \dots, T, \quad (2a)$$

$$W_t = \text{net worth in time } t, \quad i = 0, 1, \dots, t, \dots, T, \quad (2b)$$

$$B_t = \text{borrowing in time } t, \quad i = 0, 1, \dots, t, \dots, T, \quad (2c)$$

$$C_t = \text{consumption in time } t, \quad i = 0, 1, \dots, t, \dots, T, \quad (2d)$$

$$Y_t \triangleq w_t + W_t.$$

We refer to Y_t as wealth in time t . The following relationships hold:

$$B_t = C_t - Y_t, \quad (3a)$$

$$W_{t+1} = (Y_t - C_t) \cdot r. \quad (3b)$$

From (3a), positive borrowing ($B_t > 0$) represents an excess of consump-

tion over wealth. Negative borrowing constitutes investment in the safe asset.

The consumer chooses a list of consumption policies $(C_0, C_1, \dots, C_T)$ subject to the two constraints:

$$C_T = Y_T, \quad (4)$$

$$\text{Prob}\{C_0 \geq 0, C_1 \geq 0, \dots, C_T \geq 0\} = 1. \quad (5)$$

Constraint (4) is the familiar no-bequest constraint; (5) is a solvency constraint, requiring nonnegativity of consumption in each period, jointly, with probability one (w.p.1).

To state the model formally, the consumer acts so as to

$$\text{Maximize}_{\{C_0, C_1, \dots, C_T\}} E \left\{ \sum_{i=0}^T U(C_i) \delta^i \right\}, \quad \delta > 0. \quad (6)$$

Subject to Eqs. (1)–(5), E is the expectations operator taken over $w_0, w_1, \dots, w_T$. δ is one over one plus the pure rate of time discount.

III. SOLUTION OF THE MODEL

To simplify the problem, we show in Appendix A that the solvency constraint (5) implies and is implied by the inequality constraint

$$0 \leq C_t \leq Y_t + \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^i}. \quad (7)$$

In this expression $w_{\min, t+i}$ is the minimum wage in period $t+i$. In the case of random wages it makes sense to have $w_{\min, t+i} \geq 0$.

In order to use the techniques of discrete dynamic programming to solve the control problem (6), define the optimal value function

$$V^t(Y_t) = \text{Max}_{\{C_t, C_{t+1}, \dots, C_T\}} E \left\{ \sum_{i=t}^T U(C_i) \delta^{i-t} \right\} \quad (8)$$

subject to the above constraints. It is straightforward to show that $V^t(Y_t)$ is increasing, concave and thrice continuously differentiable in Y_t as long as $U(C)$ is in C . From (7), (8) and the Principle of Optimality, the following functional equation holds along the optimal path:

$$V^t(Y_t) = \text{Max}_{\{C_t\}} \{U(C_t) + \delta E V^{t+1}((Y_t - C_t) \cdot r + w_{t+1})\}, \quad (9)$$

$$\text{s.t.} \quad 0 \leq C_t \leq Y_t + \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^i}. \quad (10)$$

In order to get an informative solution to the problem, it helps if one can be sure of interior solutions in each period, i.e., interior solutions with probability one (w.p.1). In Appendix B we show that if the wage distribution is discrete, then $U'(0) = \infty$ is both necessary and sufficient for interior solutions w.p.1. If the wage distribution is continuous, then $U'(0) = \infty$ is necessary, but not sufficient. From this point on we will simply assume that we can be sure of interior solutions in each period (except the last, of course which is covered by the no-bequest constraint).

Under this assumption, the optimal level of consumption in time t , given Y_t , is such that

$$U'(C_t^*) = r\delta EV_y^{t+1}(\cdot). \quad (11)$$

In words, the current marginal utility of consumption should equal the discounted expected marginal valuation of wealth over the $T - t$ periods remaining in the planning horizon. Given the concavity of the problem (11) implies $1 > \partial C_t^* / \partial Y_t > 0$.

This in itself is rather uninformative, particularly with regard to the questions raised by the PIH. As we will see below, if we assume that the random variable ϵ_t is identically, independently distributed (i.i.d.) and that its distribution comes from a location and scale family, then we can see much more of the structure of the problem.

First, we explain the notion of a location and scale distribution. Suppose the distribution function of ϵ_t is $G(\epsilon_t)$. We first write the random variable ϵ_t in the form $\epsilon_t = l + sx_t$, which is vacuous in itself. But if the random variable x and the numbers l and s are such that $x = (w - l)/s$ has a distribution function $\psi(x_t)$ such that $\psi(x_t) = G(\epsilon_t)$ *independently of l and s* , then we say $G(\epsilon_t)$ comes from a location and scale family, which is non-trivial. Examples of location and scale families are the uniform, triangular, gamma, exponential, normal, and Cauchy. The lognormal is *not* location and scale. The location parameter is always the minimum if a minimum exists and the distribution is unbounded above. For the normal distribution it is the mean, and for the Cauchy it is the median. For distributions such as the triangular and uniform it can be the mean *or* minimum. Location and scale distributions should not be confused with the class of two-parameter distributions. Given one type of distribution with location and scale parameters l and s we generate a location and scale family by holding all parameters but l and s constant and varying l and s .

We now proceed to prove the following theorem.

THEOREM. *Assume that*

$$(A1) \quad w_t = \rho w_{t-1} + \epsilon_t,$$

(A2) ϵ_t is i.i.d. and from a location-scale family with location parameter l and scale parameter s ; $\epsilon_{\min} \leq \epsilon_t \leq \epsilon_{\max} \leq \infty$.

Define

$$H_t \triangleq W_t + w_t \cdot \sum_{i=0}^{T-t} (\rho/r)^i + l \cdot \sum_{i=1}^{T-t} \left((1/r)^i \cdot \sum_{k=0}^{i-1} \rho^k \right);$$

Then given the model, there exists a function $J^t(H_t)$ such that

$$V^t(Y_t) = J^t(H_t), \quad (\text{i})$$

and

$$C_t^* = f_t(Y_t) = g_t(H_t), \quad (\text{ii})$$

$$\frac{\partial J^t}{\partial l} = J_H^t \cdot \frac{\partial H_t}{\partial l}, \quad (\text{iii})$$

$$\frac{\partial C_t^*}{\partial l} = g_t' \cdot \frac{\partial H_t}{\partial l}. \quad (\text{iv})$$

Remark. Parts (i) and (ii) of the theorem are vacuous, since we could make l anything we wish, assume nothing about the distribution of ϵ and still write (i) and (ii) simply by substituting backwards in time. Parts (iii) and (iv) of the theorem are nontrivial and require the location and scale assumption. They assert that under (A2) if l is the location parameter of the distribution of ϵ , then it enters $C_t^* = g_t(H_t)$ and $J^t(H_t)$ only through the H_t term.

In other words it is true that we can always choose some number l such that $C_t^* = q_t(H; l)$; but if ϵ_t is location-scale and l is the location parameter, $\partial C_t^* / \partial l = g_t' \cdot (\partial H_t / \partial l)$, and similarly for $J_t(\cdot)$.

Proof. The proof is by induction. Obviously, (i)–(iv) hold for $i = T$. If we assume the theorem holds for $i = t + 1$, and can then show that it holds for $i = t$, the theorem will be proved.

Step 1. Recall condition (11):²

$$U[C_t^*] = r\delta \int_{\rho w_t + \epsilon_{\min}}^{\infty} V_y^{t+1}[(Y_t - C_t^*) \cdot r + w_{t+1}] dF_{t+1}(w_{t+1}), \quad (12)$$

where F_{t+1} is the distribution function of w_{t+1} . This defines the implicit

² We let $\epsilon_{\max} = \infty$ here, but the case where $\epsilon_{\max}$ is finite changes nothing.

function $C_t^* = f_t(Y_t)$, $1 > f'_t > 0$. Using (A1) and simply changing variables of integration, if (i) holds for $i = t + 1$

$$U'[C_t^*] = r\delta \int_{\frac{\epsilon_{\min} - l}{s}}^{\infty} J_H^{t+1} \left[(H_t - C_t^*) \cdot r + s x_{t+1} \cdot \sum_{i=0}^{T-t-1} (\rho/r)^i \right] d\psi(x_{t+1})^3. \quad (13)$$

Thus, with l fixed, $C_t^* = g_t(H_t)$, and since the L.H.S. of (12) is equal to that of (13), $C_t^* = f_t(Y_t) = g_t(H_t)$, and (ii) is proven for $i = t$.

Step 2. To prove (iv) for $i = t$, note that under (A2), $\psi(x_{t+1})$ is independent of l . Furthermore, $(\epsilon_{\min} - l)|s$ is independent of l , too; in fact, for a distribution unbounded above, $\epsilon_{\min} = l$, so $(\epsilon_{\min} - l)|s = 0$. Thus, l enters (12) only through H_t . Thus,

$$\frac{\partial C_t^*}{\partial l} = g'_t(H_t) \cdot \frac{\partial H_t}{\partial l}, \quad (14)$$

which proves (iv) for $i = t$.

Step 3. From the Principle of Optimality, we have

$$\begin{aligned} V^t(Y_t) &= U[f_t(Y_t)] + \delta \int_{\rho w_t + \epsilon_{\min}}^{\infty} V^{t+1}[(Y_t - f_t(Y_t)) \cdot r + w_{t+1}] dF_{t+1}(w_{t+1}) \\ &= U[g_t(H_t)] + \delta \int_0^{\infty} J_H^{t+1} \left[(H_t - g_t(H_t)) \cdot r + s x_{t+1} \cdot \sum_{i=0}^{T-t-1} (\rho/r)^i \right] d\psi(x_{t+1}). \end{aligned} \quad (15)$$

Since (i) holds for $i = t + 1$. But the last equality yields $J^t(H_t) = V^t(Y_t)$, which proves (i) for $i = t$.

Step 4. Given (A2) and the assumption that (iii) holds for $i = t + 1$,

$$\frac{\partial J^t}{\partial l} = \delta r \int_0^{\infty} J_H^{t+1}[\cdot] d\psi(x_{t+1}) \cdot \frac{\partial H_t}{\partial l}. \quad (16)$$

But since

$$J_{H^t}^t = \delta r \int_0^{\infty} J_H^{t+1}[\cdot] d\psi(x_{t+1}), \quad (17)$$

(iii) holds for $i = t$ and the theorem is proved.

³ The functions $J^t(H_t)$ and $V^t(Y_t)$ are equal at each point, so with constant l and s , $J_H^t = V_y^t$.

Remark 1. Although we let w_{t+1} be unbounded in the proof of the theorem, the case where $\epsilon_{t+1} \leq \epsilon_{\max} < \infty$ is easy to handle, since $(\epsilon_{\max} - l)|s$ independent of l as long as (A2) holds and l is the location parameter.

Remark 2. The usefulness of assuming ϵ_t to be location-scale depends on the question one is asking. Suppose one were to ask about the effect of adding a constant α to the wage in each time period. Independent of the distribution of ϵ_t we can create a new random variable $\epsilon'_t = \epsilon_t + \alpha$ and obtain Eq. (18) below, with α playing the role of location parameter for ϵ'_t . From this standpoint, it contributes nothing to say that ϵ_t is location-scale. But suppose that one had data on C_t , W_t , w_t , etc., and wanted to "explain" variations in consumption. If the process (1) seems reasonable in light of the data, and if the data suggest that ϵ_t comes from a location-scale distribution, then our results suggest that changes in consumption may be explained in part by changes in the location parameter, which can be estimated easily. If ϵ_t is not location-scale, then we have little to say. Thus, although this assumption is not helpful for answering questions of the first type, it could be quite useful for estimation and testing.

IV. PROPERTIES OF THE OPTIMAL SOLUTION

Using the results of the theorem and the definition of H_t , we have shown that

$$C_t^* = g_t \left\{ W_t + w_t \cdot \sum_{i=0}^{T-t} (\rho/r)^i + l \cdot \sum_{i=1}^{T-t} \left((1/r)^i \cdot \sum_{k=0}^{i-1} \rho^k \right) \right\}, \quad (18)$$

where l is the location parameter of the distribution of ϵ and enters C_t^* *only* as shown. Recall that part of the interest in C_t^* comes from whether or not it supports the PIH. Pretty clearly it does not, since the specification $w_t = \rho w_{t-1} + \epsilon_t$ ruled the PIH out from the start; however w_{t-1} got to be what it is, its value times ρ determines the distribution of w_{t+1} and all future wage distributions. The consumer does not divide the level of w_{t-1} into permanent and transitory components; all of w_{t-1} is permanent.

However, in the special case $\rho = 0$ we do get the PIH. In that case $w_t = l + sx_t$, where sx_t is random and l is not; thus the location parameter, which usually turns out to be the mean or the minimum, is a good candidate for permanent income, and sx_t is transitory income. In the case $\rho = 0$, (18) reduces to

$$C_t^* = g_t \left\{ W_t + sx_t + l \cdot \sum_{i=0}^{T-t} (1/r)^i \right\}. \quad (19)$$

Since the present value of l can be taken as the valuation of one's human capital in a world of random wages, C_t^* depends on the sum of nonhuman and human capital, plus transitory income in time t . The central feature of the PIH is borne out in this case, since with s fixed

$$\frac{\partial C_t^*/\partial l}{\partial C_t^*/\partial(sx_t)} = \sum_{i=0}^{T-t} \left(\frac{1}{r}\right)^i > 1. \quad (20)$$

In words, the marginal propensity to consume (mpc) from permanent income exceeds the mpc from transitory income, and by an amount which depends on the consumer's time horizon and the interest rate. Of course, this has always been the rationale behind the PIH (coupled with the "life cycle" approach). Our contribution here, apart from providing a rigorous derivation, is to show that under some conditions we have a precise statistical concept playing the role of permanent income, rather than an ad hoc concept such as the distributed lag on past incomes that is usually employed.

This suggests a lower bound on what the ratios of econometric estimates of the two mpc's ought to be; if $T \geq 2$, which seems reasonable, and $r = 1.10$, then the mpc from permanent income should be at least 1.9 times as large as the mpc from transitory income.

Clearly, this particular result should not be taken too seriously as an aid to econometric work, if only because wage levels are probably not i.i.d. If we allow $\rho \neq 0$, then, as we argued above, the premise behind the PIH, that the consumer divides his wage into permanent and transitory components, no longer holds. However, he does divide the *increment* to his wage over the previous period's wage into permanent and transitory components. We interpret l and sx_t as permanent and transitory *increments* to the wage, rather than as permanent and transitory *levels* of the wage.

Let us interpret Eq. (17); since $w_t = \rho w_{t-1} + l + sx_t$, we can rewrite the two power series and obtain

$$C_t^* = g_t \left\{ W_t + (\rho w_{t-1} + sx_t) \cdot \sum_{i=0}^{T-t} (\rho/r)^i + l \cdot \sum_{i=0}^{T-t} \left(\frac{\sum_{k=0}^i \rho^k}{r^i} \right) \right\}.$$

With the type of wage process assumed here, a rise in w_t with l held fixed raises the wage in period $t + i$ by an amount ρ^i , so in that sense $(\rho w_{t-1} + sx_t)$ is like an investment paying a stream of returns $\rho, \rho^2, \dots, \rho^i$. To find the present value of this investment we discount by the rate $r - 1$ and count the capitalized value as a stock of wealth available to us in the future. Now, a change in l has this effect, but also raises ϵ_{t+i} for any given sequence of realizations of the random variables $x_{t+j}, x_{t+2}, \dots, x_T$. Its cumulative impact on the wage in period $t + i$ is $\rho^i + \rho^{i-1} + \dots + \rho + 1$.

Thus, l is like an investment which pays a return $\sum_{k=0}^i \rho^k$ in period $t + i$. This stream of returns is discounted by the riskless rate and its capitalized value enters into (21). It is clear, then, that the mpc from a permanent increment to the wage is greater than the mpc from a transitory increment:

$$\frac{\partial C_t^*/\partial l}{\partial C_t^*/\partial(sx_t)} = \sum_{i=1}^{T-t} \frac{\left[\frac{\sum_{k=0}^i \rho^k}{r^i} \right]}{\sum_{i=0}^{T-t} \frac{\rho^i}{r^i}} > 1. \quad (22)$$

Thus, when $\rho \neq 0$ a modified form of the PIH holds; the consumer divides the *increment* to his wage into permanent and transitory components, and the mpc from the permanent component exceeds the mpc from the transitory component.

We should also note that even with l constant, the mpc from labor income exceeds the mpc from net worth:

$$\frac{\partial C_t^*/\partial(sx_t)}{\partial C_t^*/\partial W_t} = \sum_{i=0}^{T-t} (\rho/r)^i > 1. \quad (23)$$

The reason is that a change in W_t relaxes the current-period part of the solvency constraint, whereas a change in $w_t (= \rho w_{t-1} + \epsilon_t)$ has this effect, but also improves all future wage distributions; in terms of the solvency constraint (7), the terms $w_{\min, t+1}$ are raised. This yields theoretical insight into what is a well-known econometric result, additional to that provided by the traditional, nonstochastic life-cycle model.⁴

VI. THE EFFECTS OF INCREASED WAGE UNCERTAINTY

Three authors have concerned themselves with the effects of wage uncertainty on saving. Miller [9] solved a model with a logarithmic utility function and independent, though not identical, wage distributions, and showed that optimal consumption with uncertainty was less than without uncertainty if the wage in the latter case equaled the mean wage in the former. Yaari [14] constructed a special example in which as $T \rightarrow \infty$ optimal consumption tended to the mean wage, independently of uncertainty or the utility function. Finally, Leland [7] showed in a two-period model (in which utility was not just time-additive) that optimal saving rose or fell with increased wage uncertainty according to whether U''' was positive or negative.

⁴ See Ando and Modigliani [1].

In our case of time-additive utility we will extend Leland's result to the multiperiod case in this section of the paper. First, we define what we mean by risk. Hadar and Russell [6] and others have shown that if the utility function is increasing and concave, then given a random variable w defined on $[a, b]$ the distribution F_1 is preferred to F_2 for all such utility functions if and only if

$$\int_a^x (F_1 - F_2) dw \leq 0, \quad (24)$$

$$Vx, x \in [a, b],$$

and strict inequality holding somewhere in the interval. We say, then that an increase in risk has occurred if we start with distribution F_1 and it changes to F_2 , in the sense that our risk-averting consumer would then be worse off. A special case of this type of stochastic dominance is the mean-preserving spread analyzed by Rothschild and Stiglitz [11, 12].

Recall our first-order condition for C_t^* :

$$U'[C_t^*] = r\delta \int_{\epsilon_{\min}}^{\infty} V_y^{t+1}((Y_t + w_t(\rho/r) - C_t^*) \cdot r + \epsilon_{t+1}) dG(\epsilon_{t+1}). \quad (25)$$

Now if V_y^{t+1} were concave in Y_{t+1} , an increase in the riskiness of the distribution of ϵ_{t+1} would cause the RHS of (25) to fall, raising C_t^* ; if V_y^{t+1} were convex in Y_{t+1} , the reverse would occur. For $i = T - 1$ we have Leland's problem. In order to extend Leland's result, then, we need to show that

$$U''' \leq 0 \Rightarrow V_{yyy}^t \leq 0. \quad (26)$$

To prove this, note that since $V^T(Y_T) = U(Y_T)$, the proposition (25) holds for $i = T$. To show that it holds in general, suppose that it holds for $i = t + 1$. Now from the Principle of Optimality,

$$V^t(Y_t) = U(C_t^*) + \delta \int_{\epsilon_{\min}}^{\infty} V^{t+1}((Y_t + (\rho/r) w_t - C_t^*) \cdot r + \epsilon_{t+1}) dG(\epsilon_{t+1}). \quad (27)$$

Differentiating $V^t(Y_t)$ three times w.r.t., Y_t we have

$$V_{yyy}^t = -\delta E V_{yy}^{t+1} \cdot r^2 \cdot \frac{\partial^2 C_t^*}{\partial Y_t^2} + \delta E V_{yyy}^{t+1} \cdot \left(1 - \frac{\partial C_t^*}{\partial Y_t}\right)^2 \cdot r^3. \quad (28)$$

Obtaining $\partial C_t^*/\partial Y_t$ and $\partial^2 C_t^*/\partial Y_t^2$ from the first-order condition (25),

$$V_{yyy}^t = \frac{U''' \cdot (\delta EV_{yy}^{t+1} r^2)^3 + \delta EV_{yyy}^{t+1} \cdot r^3 (U'')^3}{(U'' + \delta EV_{yy}^{t+1} \cdot r^2)^3}. \quad (29)$$

Since U''' dictates the sign of V_{yyy}^{t+1} by assumption,

$$V_{yyy}^t \gtrless 0 \Leftarrow U''' \gtrless 0. \quad (30)$$

Q.E.D.

Thus, Leland's result extends to the multiperiod case; increased wage uncertainty raises, leaves unaffected or lowers optimal consumption according to whether U''' is negative, zero, or positive. Since any utility function with the appealing characteristic of decreasing absolute risk aversion has a positive third derivative, we have a presumption that increased wage uncertainty will cause saving to rise.

VII. CONCLUSIONS

1. Having solved the optimal consumption problem (6) with wages following the process $w_t = \rho w_{t-1} + \epsilon_t$ (and ϵ being location-scale), we obtained the strict PIH only the case $\rho = 0$; with $\rho \neq 0$ the PIH held with respect to the *increment* to the wage, but not the *level*.
2. Instead of having to rely on ad hoc concepts of permanent and transitory income, the researcher can use techniques similar to ours to determine theoretically correct and empirically estimatable exogenous variables for consumption functions.
3. We were able to extend Leland's result to the multiperiod case. Since reasonable restrictions on the risk-aversion properties of the utility function imply $U''' > 0$, our results leads one to presume that an increase in wage uncertainty will raise savings.

APPENDIX A: THE SOLVENCY CONSTRAINT

1. Let $t = T - 1$.

A. $\text{Prob}(C_T \geq 0) = 1$ implies $Y_{T-1} - C_{T-1} \geq -w_{\min,T}/r$.

Proof.

$$C_T = W_T + w_T = (Y_{T-1} - C_{T-1}) \cdot r + w_T.$$

For $C_T \geq 0$ w.p.1, we require $(Y_{T-1} - C_{T-1}) \cdot r + w_{\min,T} \geq 0$, or $(Y_{T-1} - C_{T-1}) \cdot r \geq -w_{\min,T}$.

B. $(Y_{T-1} - C_{T-1}) \geq -w_{\min,T}/r$ implies $\text{Prob}(C_T \geq 0) = 1$.

2. Let $t = T - 2$.

A. $\text{Prob}(C_{T-1} \geq 0, C_T \geq 0) = 1$ implies

$$Y_{T-2} - C_{T-2} \geq -\left(\frac{w_{\min,T-1}}{r} + \frac{w_{\min,T}}{r^2}\right).$$

Proof.

For $C_{T-1}, C_T \geq 0$ we need:

$$0 \leq C_{T-1} \leq Y_{T-1} + \frac{w_{\min,T}}{r}.$$

So for $C_{T-1}, C_T \geq 0$, w.p.1, we have

$$\begin{aligned} 0 &\leq W_{T-1} + w_{\min,T-1} + \frac{w_{\min,T}}{r} \\ &= (Y_{T-2} - C_{T-2}) \cdot r + w_{\min,T-1} + \frac{w_{\min,T}}{r}, \end{aligned}$$

or

$$(Y_{T-2} - C_{T-2}) \geq -\left(\frac{w_{\min,T-1}}{r} + \frac{w_{\min,T}}{r^2}\right).$$

B. $Y_{T-2} - C_{T-2} \geq -(w_{\min,T-1}/r + w_{\min,T}/r^2)$ implies $\text{Prob}(C_{T-1} \geq 0, C_T \geq 0) = 1$.

Proof.

Suppose $Y_{T-2} - C_{T-2} < -(w_{\min,T-1}/r + w_{\min,T}/r^2)$; then if $w_{T-1} = w_{\min,T-1}$,

$$C_{T-1} \leq W_{T-1} + w_{\min,T-1} + \frac{w_{\min,T}}{r} < 0.$$

Suppose $Y_{T-2} - C_{T-2} \geq -(w_{\min,T-1}/r + w_{\min,T}/r^2)$; then if $w_{T-1} = w_{\min,T-1}$,

$$\begin{aligned} W_{T-1} + w_{\min,T-1} + \frac{w_{\min,T}}{r} \\ \geq -\left(\frac{w_{\min,T-1}}{r} + \frac{w_{\min,T}}{r^2}\right) \cdot r + w_{\min,T-1} + \frac{w_{\min,T}}{r} > 0, \end{aligned}$$

so

$$C_{T-1} \geq 0, \quad \text{w.p.1.}$$

3. The same demonstration can be made recursively back to any $t < T$.

APPENDIX B: INTERIOR SOLUTIONS

We wish to show that if the distribution of the wage is discrete, then $U'(0) = \infty$ is both necessary and sufficient for interior solutions w.p.1. We show sufficiency first. Proposition. If $U'(0) = \infty$ and the distribution w_t is discrete, then

$$\left. \begin{aligned} V_y^t \\ Y_t \end{aligned} \right| = \infty, \quad Y_t = - \sum_{i=1}^{T-t} \left(\frac{w_{\min, t+i}}{r^i} \right).$$

1. For $i = T$ the statement holds, obviously.
2. Assume the proposition holds for $t + 1$, i.e.,

$$\left. \begin{aligned} V^{t+1} \\ Y_{t+1} \end{aligned} \right| = \infty, \quad Y_{t+1} = - \sum_{i=0}^{T-t-1} \frac{w_{\min, t+i}}{r^i}.$$

From the Principle of Optimality,

$$V^t(Y_t) = \text{Max}_{\{C_t\}} \{U(C_t) + \delta EV^{t+1}((Y_t - C_t) \cdot r + w_{t+1})\},$$

$$\text{s.t.} \quad Y_t - C_t \geq - \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^i},$$

and

$$U'(C_t^*) = \delta r EV_y^{t+1}.$$

Suppose $Y_t - C_t = - \sum_{i=1}^{T-t} (w_{\min, t+i}/r^i)$; then in period $t + 1$, if

$$w_{t+1} = w_{\min, t+1}, \quad Y_{t+1} = - \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^{i-1}} + w_{\min, t+1} = - \sum_{i=1}^{T-t-1} \frac{w_{\min, t+i+1}}{r^i}.$$

Since $V_y^{t+1} = \infty$ at that point by assumption, and since $\text{Prob}(w_{t+1} = w_{\min, t+1}) = \pi_{\min} > 0$,

$$\left. \begin{aligned} \pi_{\min} \cdot V_y^{t+1} \\ Y_{t+1} \end{aligned} \right| = \infty, \quad Y_{t+1} = - \sum_{i=1}^{T-t-1} \frac{w_{\min, t+i}}{r^i},$$

and $V_y^t = \infty$ evaluated at $Y_t - C_t = - \sum_{i=1}^{T-t} (w_{\min, t+i})/r^i$.

3. Since the proposition holds for $i = T$, and since to assume that it holds for $i = t + 1$ means that it is true for $i = t$, then proposition holds for all i .

Now, define

$$\phi^t(Y_t) \triangleq U(C_t) + \delta EV^{t+1}((Y_t - C_t) \cdot r + w_{t+1}),$$

and consider

$$\left. \begin{aligned} \frac{\partial \phi^t}{\partial C_t} &= U'(Y_t + \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^i}) - r \delta EV_y^{t+1} \left(-r \cdot \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^i} + w_{t+1} \right), \\ C_t &= Y_t + \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^i}. \end{aligned} \right\}$$

But as we showed in the previous proposition, this last RHS is infinitely negative; then,

$$C_t < Y_t + \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^i}.$$

At the lower bound, $C_t = 0$, $U'(0) = \infty$ makes $\partial \phi^t / \partial C_t = \infty$, so $C_t > 0$ at the optimum. Putting these results together, we have

$$0 < C_t < Y_t + \sum_{i=1}^{T-t} \frac{w_{\min, t+i}}{r^i}$$

at the optimum.

All of this presumed of course, that $C_t > 0$ was feasible. This boils down to a condition on initial net worth, extending our results on the solvency constraint back to $i = 0$, we have

$$\text{Prob}\{C_0 \geq 0, C_1 \geq 0, \dots, C_T \geq 0\} = 1,$$

iff

$$Y_0 \geq - \sum_{i=1}^T \frac{w_{\min, i}}{r^i} \equiv \hat{Y}_0.$$

Thus, if $Y_0 > \hat{Y}_0$, and the wage distribution is discrete, then $U'(0) = \infty$ ensures $C_t > 0$ and future solvency with interior solutions w.p.1.

To see how continuous wage distributions make things less clear, let $t = T - 1$; then V_y^T is simply U' and

$$EU'(-w_{\min, T} + w_T)$$

is not necessarily infinite. If the density function of w tends to zero as $w_T \rightarrow w_{\min, T}$ more rapidly than U' tends to infinity, then

$$EU'(-w_{\min, t} + w_T)$$

exists and we could have C_{T-1} at its upper bound. Thus interior solutions do not exist w.p.l. for continuous wage distributions in general.

To show sufficiency, merely note that without $U'(0) = \infty$ we could have $C_{T-1}^* = 0$ merely by letting r get very large.

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